Student Marks Prediction Using Machine Learning

1. Introduction

Machine Learning is a branch of Artificial Intelligence that enables systems to learn from data and make predictions. This project predicts student marks based on study hours using Linear Regression.

2. Objective

To build a machine learning model that predicts student marks based on the number of study hours.

3. Technologies Used

- Python
- Google Colab
- Pandas
- Scikit-learn
- Matplotlib

4. Dataset

The Student Scores dataset contains study hours and corresponding student marks. The dataset is used to train the Linear Regression model.

5. Algorithm Used

Linear Regression algorithm is used to predict marks based on study hours.

6. Procedure

1. Import required libraries.
2. Load dataset using pandas.
3. Split dataset into training and testing data.
4. Train Linear Regression model.
5. Predict marks.
6. Evaluate model performance.
7. Display graph visualization.

7. Output

The model successfully predicts student marks and provides graphical visualization of the prediction line.

8. Result

The project achieved a high R2 Score and low Mean Absolute Error, indicating good prediction accuracy.

9. Conclusion

Machine Learning models can effectively predict student performance using study-hour data. Linear Regression provides accurate and simple prediction results.